

Lab Worksheet 2: Numerical Data

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GitHub Repository: <https://github.com/Dulani-Creator/CT2022062-LW2>

Question 1: Math Expressions

Code:

```
1  package Q1;
2
3  public class main {
4      public static void main(String[] args) {
5
6          double A = 2, B = 5, C = 1;
7          double X = 10, Y = 3;
8          double radius = 7;
9
10
11         double resultA = Math.sqrt(Math.pow(B, 2) + 4 * A * C);
12         System.out.println("Result of a: " + resultA);
13
14
15         double resultB = Math.sqrt(X + 4 * Math.pow(Y, 3));
16         System.out.println("Result of b: " + resultB);
17
18
19         double resultC = Math.cbrt(X * Y);
20         System.out.println("Result of c: " + resultC);
21
22
23         double area = Math.PI * Math.pow(radius, 2);
24         System.out.println("Area of the circle: " + area);
25     }
26 }
```

Question 2: Centimeters to Feet and Inches

Code:

```
1 package Q2;
2 import java.util.Scanner;
3 public class main {
4     public static void main(String[] args) {
5         Scanner scanner = new Scanner(System.in);
6
7         System.out.print("Enter length in centimeters: ");
8         double cm = scanner.nextDouble();
9
10
11         double totalInches = cm / 2.54;
12
13
14         int feet = (int) (totalInches / 12);
15
16
17         double remainingInches = totalInches % 12;
18
19         System.out.printf("%.2f cm is %d feet and %.2f inches.\n", cm, feet, remainingInches);
20
21         scanner.close();
22     }
23 }
24
```

Question 3: Celsius to Fahrenheit

Code:

```
1 package Q3;
2 import java.util.Scanner;
3 public class main {
4     public static void main(String[] args) {
5         Scanner scanner = new Scanner(System.in);
6
7         System.out.print("Enter temperature in degrees Celsius: ");
8         double celsius = scanner.nextDouble();
9
10
11         double fahrenheit = (1.8 * celsius) + 32;
12
13         System.out.println("Temperature in Fahrenheit: " + fahrenheit);
14
15         scanner.close();
16     }
17 }
```

Question 4: Calorie Calculator

Code:

```
1 package Q4;
2 import java.util.Scanner;
3 public class main {
4     public static void main(String[] args) {
5         Scanner scanner = new Scanner(System.in);
6
7         System.out.print("Enter your weight in pounds: ");
8         double weight = scanner.nextDouble();
9
10
11         double calories = weight * 19;
12
13         System.out.println("You need " + calories + " calories in one day.");
14
15         scanner.close();
16     }
17 }
```

Question 5: Fahrenheit to Celsius

Code:

```
1 package Q5;
2 import java.util.Scanner;
3 public class main {
4     public static void main(String[] args) {
5         Scanner scanner = new Scanner(System.in);
6
7         System.out.print("Enter temperature in degrees Fahrenheit: ");
8         double fahrenheit = scanner.nextDouble();
9
10         double celsius = (5.0 / 9.0) * (fahrenheit - 32);
11
12         System.out.printf("Temperature in Celsius: %.2f\n", celsius);
13
14         scanner.close();
15     }
16 }
```

Question 6: Age Calculator

Code:

```
1 package Q6;
2 import java.util.Scanner;
3 import java.time.Year;
4 public class main {
5     public static void main(String[] args) {
6         Scanner scanner = new Scanner(System.in);
7
8         System.out.print("Enter the year you were born: ");
9         int birthYear = scanner.nextInt();
10
11         int currentYear = Year.now().getValue();
12         int age = currentYear - birthYear;
13
14         System.out.println("You were born in " + birthYear + " and will be (are) " + age + " this year.");
15
16         scanner.close();
17     }
18 }
```

Question 7: BMI Calculator

Code:

```
1 package Q7;
2 import java.util.Scanner;
3 public class main {
4     public static void main(String[] args) {
5         Scanner scanner = new Scanner(System.in);
6
7         System.out.print("Enter weight in kilograms: ");
8         int weight = scanner.nextInt();
9
10        System.out.print("Enter height in centimeters: ");
11        int height = scanner.nextInt();
12
13        double bmi = weight / Math.pow(height / 100.0, 2);
14
15        System.out.printf("Your BMI is: %.2f\n", bmi);
16
17        scanner.close();
18    }
19 }
```

Question 8: Volume of a Sphere

Code:

```
1 package Q8;
2 import java.util.Scanner;
3 public class main {
4     public static void main(String[] args) {
5         Scanner scanner = new Scanner(System.in);
6
7         System.out.print("Enter the radius of the sphere (r): ");
8         double r = scanner.nextDouble();
9
10        double volume = (4.0 / 3.0) * (Math.PI * Math.pow(r, 3));
11
12        System.out.printf("The volume of the sphere is: %.2f\n", volume);
13
14        scanner.close();
15    }
16 }
```

Question 9: Investment Calculator

Code:

```
1 package Q9;
2 import java.util.Scanner;
3 public class main {
4     public static void main(String[] args) {
5         Scanner scanner = new Scanner(System.in);
6
7         System.out.print("Enter investment amount in dollars (P): ");
8         double p = scanner.nextDouble();
9
10        System.out.print("Enter interest rate percentage (R): ");
11        double r = scanner.nextDouble();
12
13        System.out.print("Enter number of years (N): ");
14        int n = scanner.nextInt();
15
16        double amount = p * Math.pow(1 + (r / 100.0), n);
17
18        System.out.printf("Amount of money earned after %d years: $%.2f\n", n, amount);
19
20        scanner.close();
21    }
22 }
```

Question 10: Loan Calculator

Code:

```
1  package Q10;
2  import java.util.Scanner;
3  public class main {
4      public static void main(String[] args) {
5          Scanner scanner = new Scanner(System.in);
6          final int MONTHS_IN_YEAR = 12;
7
8          System.out.print("Enter loan amount: ");
9          double loanAmount = scanner.nextDouble();
10
11         System.out.print("Enter annual interest rate (e.g., 5.5): ");
12         double annualInterestRate = scanner.nextDouble();
13
14         System.out.print("Enter loan period in years: ");
15         int loanPeriod = scanner.nextInt();
16
17         double monthlyInterestRate = annualInterestRate / 100.0 / MONTHS_IN_YEAR;
18
19         int numberOfPayments = loanPeriod * MONTHS_IN_YEAR;
20
21         double monthlyPayment = (loanAmount * monthlyInterestRate) /
22             (1 - Math.pow(1 / (1 + monthlyInterestRate), numberOfPayments));
23
24         double totalPayment = monthlyPayment * numberOfPayments;
25
26         System.out.printf("Monthly Payment: %.2f\n", monthlyPayment);
27         System.out.printf("Total Payment: %.2f\n", totalPayment);
28
29         scanner.close();
30     }
31 }
32
```